

Review

Overview of Existing Multi-Criteria Decision-Making (MCDM) Methods Used in Industrial Environments

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Abstract

The selection of an appropriate technological process is essential to achieve optimal results in manufacturing companies. This affects quality, efficiency and competitiveness. In the modern industry, multi-criteria decision-making (MCDM) methods are increasingly used to evaluate, optimize and solve various manufacturing challenges. In this review article, existing methodologies and patents related to optimization and decision making are investigated. The main characteristics and applications of the methods are outlined. The purpose of this article is to provide a systematic review and evaluation of the main MCDM methods used in industrial practice, including through an analysis of relevant methodologies and patents. The methodology involves a structured literature and patent review, focusing on applications of widely used MCDM techniques such as the AHP (analytic hierarchy process), ANP (analytic network process), FUCOM (full consistency method), TOPSIS (technique for order preference by similarity to ideal solution), and VIKOR (višekriterijumsko kompromisno rangiranje). The analysis outlines each method's strengths, limitations and areas of applicability. Special attention is given to the potential of the FUCOM for process evaluation in manufacturing. The findings are intended to guide researchers and practitioners in selecting appropriate decision-making tools based on specific industrial contexts and objectives. In conclusion, from the comparative analysis made, the methodologies reveal their advantages and disadvantages as well as limitations that arise in their application.

Keywords: MCDM; AHP; FUCOM; VIKOR; TOPSIS; ANP; BWM; hybrid MCDM; methodology; fuzzy



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1. Introduction

In today's increasingly complex and dynamic markets, companies are challenged to optimize their processes to meet the demands for high quality, low cost and fast production. In a manufacturing environment, professionals are faced with the need to make multiple decisions on a daily basis related to a variety of engineering tasks ranging from strategic planning of manufacturing operations to the effective management of process flows [1]. In manufacturing environments, the selection of a rational technological process is of great importance to ensure the quality of the final product as well as the competitiveness of the company. With modernization and increasing market demands, this choice is becoming

more and more complex due to the need to balance between multiple criteria such as cost, lead time and product quality.

In order to consider the importance of each of these criteria and to make an informed decision, multi-criteria decision-making (MCDM) methods provide reliable and different methods to assist in the selection. In modern manufacturing practice, these methods have become an important part because of their ability to strike a balance between multiple and sometimes conflicting criteria [2].

Selecting the most suitable criteria that define a rational technological process requires a deep understanding of the specific needs of production. This includes an analysis of the materials used, the technical requirements of the final product, the limitations of the production equipment and the requirements of the customers.

The flexibility of production processes influences the implementation of different methodologies in real conditions. With the dynamic changes and conditions of customers and the market, the modernization in technological processes and the integration of Industry 4.0., companies have to lead a continuous race with their adaptation to these conditions. Delaying the integration of the production process into these conditions may lead to incorrect technological process selections or to making gaps in their development, resulting in low-quality production. Ignoring this aspect may damage the reputation of the manufacturer and put its competitiveness in doubt [3,4].

MCDM methods evaluate, rank and assign importance to set criteria; in this way they provide a reliable and structured approach to solving a specific problem or task. In this way, professionals can make rational and justified decisions that reflect the priorities set at the beginning.

In recent years, multiple MCDM methods such as the analytic hierarchy process (AHP), fuzzy AHP (FAHP), analytic network process (ANP), technique for order preference by similarity to ideal solution (TOPSIS), višekriterijumsko kompromisno rangiranje (VIKOR), etc., have been more frequently applied. They have an increasingly wide application in manufacturing practice—for example, in the selection of machines, tools, materials and other resources. The aim of which is to increase production efficiency and achieve more sustainable processes.

The selection of literature sources and patents included in this review was guided by their relevance and relevance to MCDM methods applied in manufacturing and processing industries. Publications were identified through databases such as Scopus, Web of Science and Google Scholar, using keywords including “MCDM”, “AHP”, “FUCOM”, “process optimization”, “manufacturing systems”, etc. Priority was given to peer-reviewed articles published in the last 10 years, as well as widely cited fundamental papers and patents with industrial applications. Duplicate or irrelevant sources were excluded. The final selection comprised over 200 sources, ensuring both the depth and diversity of methodologies and applications.

In view of the collected literature and identified research areas, the review is organized into several main steps, as shown in Figure 1.

- Step 1—Overview of the existing situation: This covers the collection of data and information on existing MCDM methods, methodologies and patents that implement these methods;
- Step 2—Analysis of the considered methods: Similarities and differences between the different methods are analyzed, as well as their advantages and disadvantages;
- Step 3—Challenges and future directions: The main conclusions are summarized and suggestions for future research and development in the field of MCDM are formulated, with a focus on their application in technological process optimization.

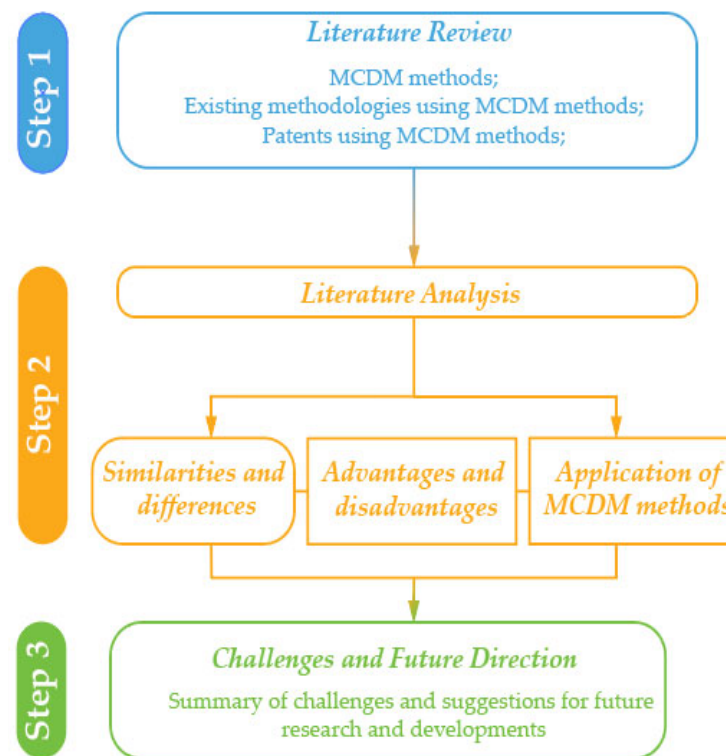


Figure 1. Visual flow of the literature review of the paper.

This paper takes a closer look at some recent methods and patents that try to apply MCDM techniques in manufacturing. The focus is on how these approaches are used in different industries and what makes them work (or not work) in those settings.

The paper goes through several examples and points out where the methods are similar and where they differ.

The analysis also examines the methodologies employed by decision makers to determine criteria weights, how they collect and handle the data, and how MCDM methods are actually used. The objective is not just to compare them but to obtain a better feel for when each method makes sense—and when it does not. Its findings would be valuable for both the academic community and professionals interested in the development of this field in order to develop new methods and tools to improve manufacturing practices.

2. Materials and Methods

2.1. Multi-Criteria Decision Making (MCDM)

MCDM methods represent a structured set of analytical techniques designed to support informed decision making in complex or uncertain environments. They facilitate evaluation and selection among multiple alternatives (e.g., manufacturing processes, equipment, materials, etc.) based on diverse criteria, which can often overlap, conflict or be very similar in importance. These criteria may include both quantitative and qualitative aspects such as quality, cost, time, sustainability and efficiency, often measured in different units. In manufacturing, MCDM methods are particularly valuable due to their ability to systematically balance such multiple and conflicting factors [5–8].

Some of the first MCDM methods emerged in the mid-20th century, and a variety of techniques have been developed over the years. New MCDM methods are developed as a consequence of a specific problem (practical or not) involving multi-criteria problems such as optimization of production processes, cost-effective resource management, adaptation to changing conditions, etc. By studying their past, the progression over time and continuous

improvement in methods to the changing dynamic environment can be analyzed. Figure 2 traces the development of multi-criteria MCDM methods from the mid-20th century to the present day. The timeline highlights several moments, including the emergence of the AHP in the 1980s, its expansion into the ANP, and the introduction of newer methods such as the FUCOM and BWM. This visual representation helps to see how new methods build upon previous ones and how they adapt to the needs of industry.

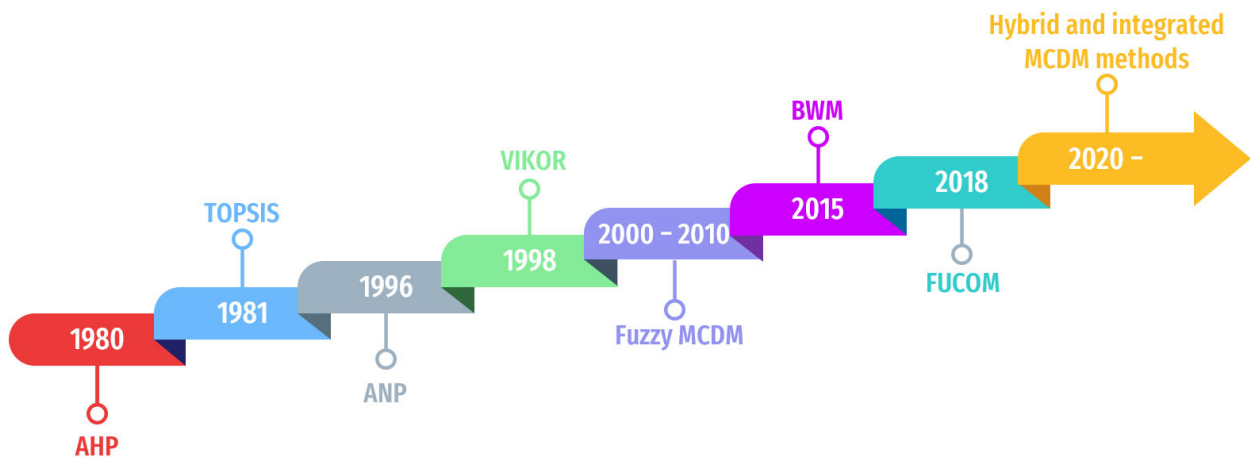


Figure 2. Timeline of MCDM methods.

Some of the most commonly used MCDM methods are presented in Figure 3. They provide opportunities that allow experts to objectively and systematically evaluate different alternatives, thereby supporting optimal decision making in complex manufacturing environments.

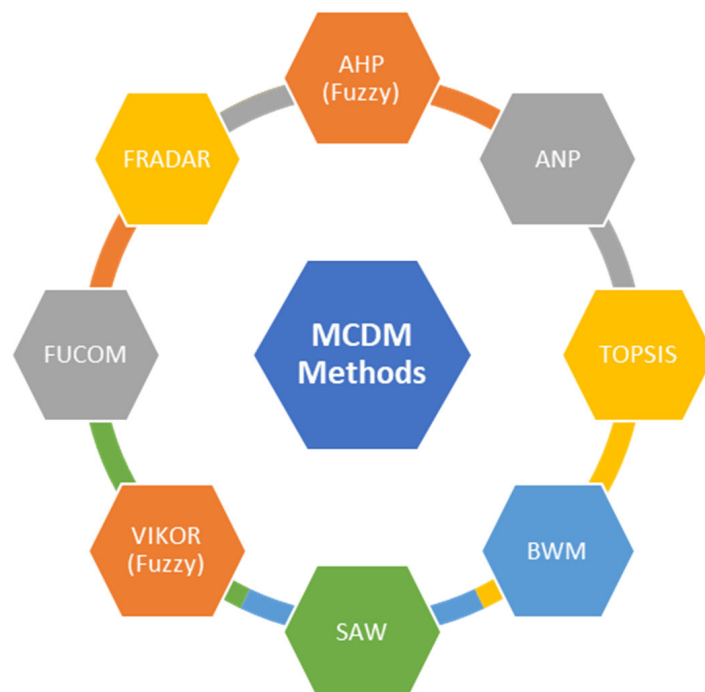


Figure 3. Commonly used MCDM methods.

The selected eight MCDM methods were chosen based on their widespread use in recent manufacturing literature and their coverage in recent comparative methodological studies. These methods represent a balance between traditional (e.g., AHP), advanced (e.g., ANP, FUCOM) and computationally efficient (e.g., BWM) approaches.

- Analytic Hierarchy Process (AHP):

The AHP is one of the first and most widely used MCDM methods, developed in the 1980s. The method structures complex decisions by building a hierarchical structure in which the main goal is broken down into sub-goals, criteria and sub-criteria. The AHP method applies pairwise comparisons between criteria, calculating the relative weight of each and evaluating different variances (alternatives) based on the resulting weights [9,10]. The method is used for solving complex problems, the foundation of which is clear and consistent organization that supports the selection of the optimal variant through systematic analysis. The MCDM method, the AHP, allows the exploration of different (diverse) criteria and the verification of logical consistency, thus simplifying informed and justifiable decision making. This would be particularly useful in situations where a precise balance between the different criteria is required.

- Fuzzy AHP (FAHP):

The FAHP is an MCDM method that combines the traditional AHP with fuzzy logic. This method allows for the handling of uncertainty and subjectivity in expert estimates by using fuzzy numbers instead of exact values. This is particularly useful when expert opinions are unclear or when criteria cannot be accurately and quantitatively assessed [11].

By using degrees of membership, the FAHP method allows a more flexible approach to expressing preferences. This advantage of dealing with the “uncertainty” (ambiguity) in expert opinions makes it an extremely subjective method. In traditional logic, an element belongs or does not belong to the set. In the FAHP, fuzzy logic is used, which iterates with numerical values between 0 and 1, allowing for a more flexible reflection or approximation to reality. In this way, expert opinions are represented more realistically, and the influence of subjectivity is reduced. Despite its complexity, the method has wide applicability in various fields such as business, engineering, risk management, etc., because of its accurate and reliable results [12].

- Analytic Network Process (ANP):

The ANP is an extension of the AHP method. While AHP models decision problems through a hierarchical structure with unidirectional relationships, the ANP allows for more complex interactions through a network structure where elements can have mutual dependencies and inverse relationships [10].

The traditional AHP is limited in modeling such interdependencies as it assumes independence between levels of the hierarchy and between elements at a given level. The ANP overcomes these limitations by implementing a network structure that allows complex dependencies and feedback between different elements to be modeled [13]. The main idea of the ANP is to capture the mutual influence between criteria and alternatives using a super matrix that represents all possible dependencies. By raising the super matrix to the degree necessary to reach convergence, we can obtain stable item priorities. This method allows for a more realistic and accurate modeling of complex systems where elements influence each other in different ways [14].

- Technique for Order Preference by Similarity to Ideal Solution (TOPSIS):

The TOPSIS method is one of the most widely used MCDM methods. It is based on the concept that the best alternative is the one that is closest to the ideal solution (positive ideal solution, PIS) and farthest from the non-ideal solution (negative ideal solution, NIS) [15,16]. This method is characterized by its simplicity, logical clarity and ability to handle a large number of criteria and alternatives. An ideal solution represents a hypothetical alternative that has the best possible values for all criteria. This means maximizing the useful (favorable) criteria and minimizing the unfavorable (undesirable)

criteria. A non-ideal solution is a hypothetical alternative that has the worst possible values for all criteria [17]. In manufacturing environments, the TOPSIS is used to select optimal manufacturing equipment, materials and processes based on multiple criteria including cost, performance and sustainability. The method has been successfully applied to the evaluation of transportation and logistics systems in manufacturing [18,19].

- Best Worst Method (BWM):

The BWM is an efficient and innovative method for determining the weights of criteria in MCDM. By reducing the number of comparisons required and increasing the consistency of scores, the BWM facilitates the decision-making process and improves the reliability of the results [20]. Despite some limitations, the method is widely applicable in various fields and continues to evolve through new extensions and combinations with other techniques [21].

- Simple Additive Weighting (SAW) Method:

The SAW method is also known as linear weighted summation and is preferred for its simplicity, intuitiveness and ease of application in a variety of fields. This method is based on evaluating alternatives by adding up their weighted values according to different criteria [22]. Through the assumption of linear compensation between criteria, i.e., lower scores on a criterion can be compensated for by other higher scores, which is the main basis of the SAW method, SAW is particularly suitable for situations where the criteria are independent and quantitatively measurable [23].

- Višekriterijumsko Kompromisno Rangiranje (VIKOR):

As the complexity of multi-criteria problems grows, the need for new MCDM methods increases. This is where VIKOR is designed to make the complexity of tasks easier and faster. It is based on compromise solution theory, thus balancing among conflicting criteria. By providing a structured approach to evaluate and rank the criteria under consideration, it is able to quickly derive a solution to the problems under consideration [24,25]. VIKOR is suitable in cases where the expert desires a solution that most closely approximates the ideal outcome.

- Full Consistency Method (FUCOM):

The FUCOM is an accurate and efficient MCDM method for determining criteria weights. By minimizing biases and ensuring complete consistency of expert evaluations, the FUCOM provides reliable results, significantly improving the quality of decision making [26–30]. Despite the need for precise ordering of criteria and its dependence on expert scores, the advantages of the method make it a valuable tool in areas such as logistics, risk management and sustainable development.

Figure 4 shows the sequence of steps in applying the method. Following this sequence provides a logical, systematic approach and high reliability and consistency in the decision-making process.

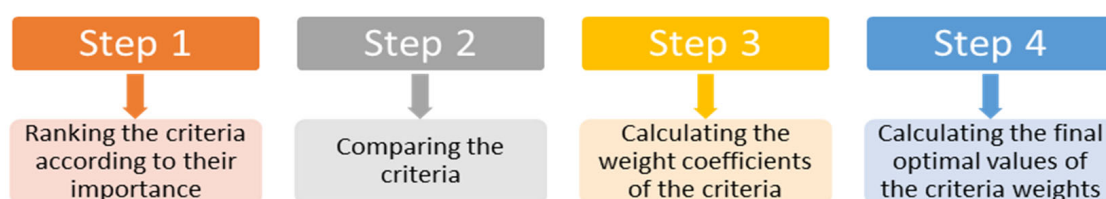


Figure 4. Step-by-step implementation of the MCDM method—the FUCOM.

- Fuzzy RADAR (FRADAR).

FRADAR is an MCDM method integrating fuzzy logic to enhance the process failure mode and effects analysis (PFMEA). It addresses uncertainties in expert judgments using fuzzy sets and evaluates criteria such as action priority (AP), risk priority number (RPN), cost effectiveness, resolution time and production impacts. FRADAR improves decision accuracy, which is particularly valuable for automotive industry applications requiring precise risk assessment and failure prioritization. The method can also be extended by using interval-valued Pythagorean fuzzy numbers (IPF-RADAR), offering deeper flexibility and accuracy in evaluating team composition and candidate skills, enhancing objectivity in multidisciplinary risk management scenarios [31,32].

These days, sustainable production is receiving more attention. It is something many industries are starting to focus on, especially with all the changes happening around us. Because of that, MCDM methods are being used more often. They help people deal with complicated problems by looking at several criteria, not just one. This way of thinking makes it easier to choose between options. It also helps meet different standards and follow specific rules more smoothly.

Good knowledge of the characteristics, advantages and limitations of the different MCDM methods allows practitioners to select the most appropriate tool according to the specific case. Through careful analysis and appropriate application of the methods, organizations and individual experts can make more informed, justified and effective decisions that meet their objectives and priorities. The level of expertise, available resources and the specificity of the problem are important factors to consider when using MCDM methods. The adaptation of different methods to specific conditions, as well as the combination between them, leads to robust and reliable results. MCDM methods offer a variety of approaches—from classic hierarchical models to modern techniques optimized for fast and reliable results. Their applications in various industries should be examined in order to assess their real contribution to the optimization of technological processes.

2.2. MCDM Methods in Different Manufacturing Industries

Now that the main methods have been outlined, the next step is to look at their application in the field. MCDM methods are applied in various industries, where they simplify complex decision-making problems and support process optimization. Various complex tasks arise during the workflow, and the main role of these methods is to help with solutions. With their assistance, specialists are able to compare different alternatives with multiple criteria of varying importance such as cost, quality, lead time, etc. They represent a significant contribution to the development of modern economics with their use in different industries [33–39].

In the field of sustainable engineering, complex situations are often encountered in which a balance between economic benefits, environmental impact, social aspects and engineering cases needs to be achieved. According to the authors of [40], MCDM methods are particularly useful in optimizing engineering projects in major industries such as construction and infrastructure, energy, transportation, logistics and supply chain management.

At the core are MCDM methods, which bring together some of the key stages in manufacturing: process and supply management, resource management, quality control and risk management, as well as product development and innovation. Figure 5 presents the different manufacturing industries using MCDM methods, brought together from the different stages of production. Various industrial sectors are shown—from automotive and electronics to chemical and energy industries. The connections emphasize that these methods are not limited to a single field but have a wide scope. This proves their adaptability and applicability in solving a variety of engineering and management problems.

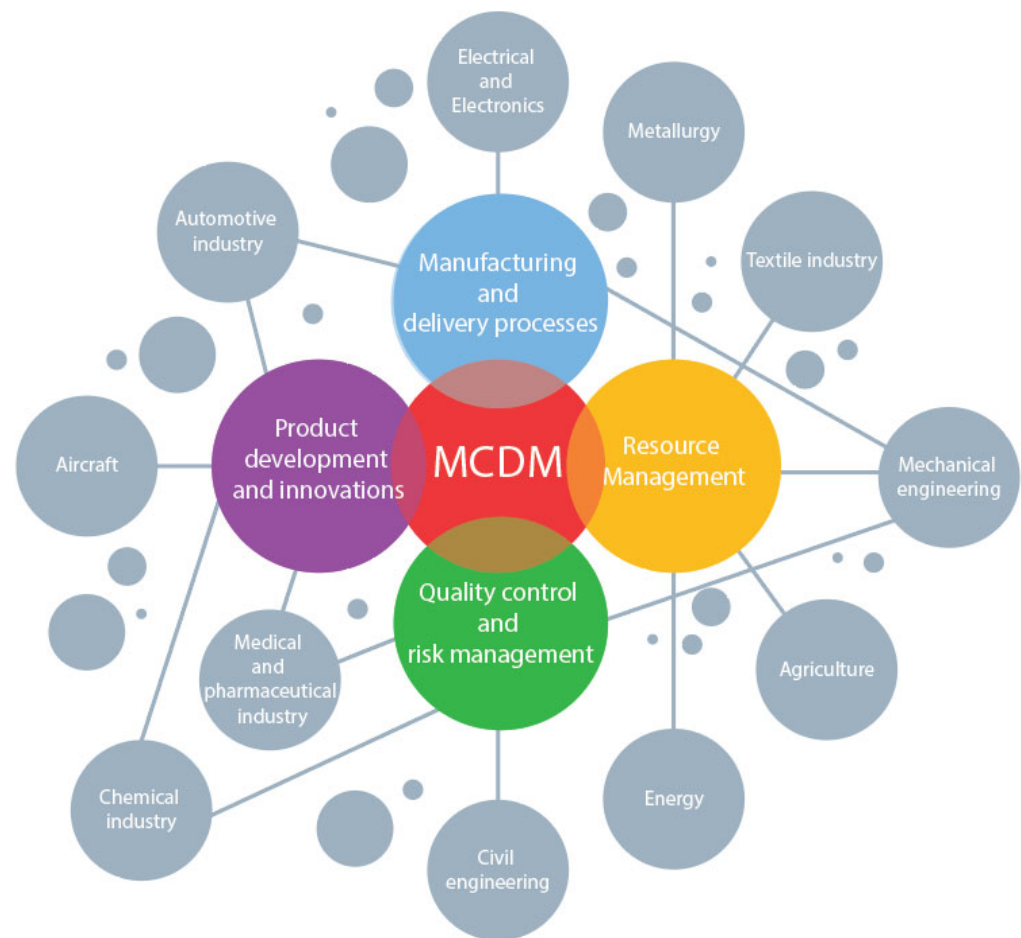


Figure 5. The connections between different manufacturing industries using MCDM methods.

In modern industries, manufacturing is still of critical importance. It is tied to how companies stay efficient, how they stay in the game, and whether their products are actually good enough. People working in this area deal with all sorts of things—supply chains, figuring out the right technologies and picking the right materials. And when things become complicated, which they often do, MCDM methods come in handy. They allow you to look at the problem from different perspectives. Multiple variables are considered simultaneously, enabling a more comprehensive evaluation. While the approach has limitations, it facilitates more robust and potentially sustainable decision-making processes.

An example of this is the automotive industry, where MCDM methods are used to select suitable suppliers (offering the best price/quality/delivery time ratio) and to select materials for individual vehicle components [41–49].

In another industrial sector, such as electronics and electrical engineering, MCDM methods are used to manage complex supply chains, which often involve international partners [50].

Mechanical engineering, as the backbone of industrial manufacturing, often faces challenges related to the selection of technologies, materials and processes [51–56]. Authors of publications on the subject [57,58] consider MCDM methods to support process optimization such as turning, milling, drilling, etc. Other authors [59–67] highlight that incorrect machine selection can lead to increased costs, reduced efficiency and even loss of competitiveness. The analysis of different MCDM methods such as the AHP, the TOPSIS, VIKOR, etc., shows that the most commonly used criteria are technical and economic.

MCDM methods in the chemical industry are used to select optimal catalysts, select raw materials and engineer energy efficient processes. According to the authors of [68], one

of the main challenges of MCDM in chemical engineering is the complexity of the processes, which requires the integration of different methods for normalization, criteria weighting and analysis of the results. In order to reduce the production of hazardous waste and to aid in resource recovery and environmental protection, the authors of [69] conducted a study using MCDM methods in the healthcare sector, specifically for hazardous waste disposal.

For companies to minimize waste and aim for sustainability, effective management of their resources is essential. For choosing among alternatives in the use of natural, energy and material resources, MCDM methods and their approaches provide a basis for making and evaluating exactly these choices, in order to achieve a balance for both economic and environmental purposes [70–72]. Resource management is particularly important for metallurgy, the textile industry, the energy sector, engineering and agriculture [73–77]. The industries listed so far and the companies in them that apply MCDM methods to optimize energy efficiency, select sustainable materials and manage water resources, with the primary goal of utilizing natural resources responsibly, maximally and sustainably.

Quality control and risk management are essential to ensure product safety, reliability and compliance with set standard requirements. For this purpose, analytical tools based on MCDM methods have been developed to assess risks and help implement targeted quality improvement strategies. These include the aerospace, chemical and construction industries [78–80]. In aviation, these methods support the assessment of risks and component reliability, while in construction they are used to select materials and techniques that ensure safety and durability [81].

According to the authors of [82], the application of MCDM methods (LBWA and SAW) is focused on the field of information technology and system analysis, where they are used to select appropriate standards for modeling business processes in organizations. The methods are applied to compare and rank the most commonly used standards (DFD, IDEF0, IDEF3, and BPMN), allowing system analysts and management to determine the most appropriate standard according to the specific criteria and needs of the particular organization, supporting effective decision making and process optimization.

In product development and innovation, in recent years, quite a few MCDM methods have been implemented to support strategic decisions related to the selection of new technologies, materials and features, allowing companies to create competitive products and meet the demands of dynamically changing markets.

2.3. Existing Methodologies Using MCDM Methods for Technological Process Evaluation

In technological process development, especially when carried out by beginner technologists with insufficient practical experience, it is common to see the development of different technological processes that can lead to incorrect decisions. These can negatively affect the selection of a rational technological process by focusing on less important features of the part instead of focusing on key aspects. A number of experts and researchers from different fields continue to study and examine this trend.

The complexity of process design and the need to balance between multiple criteria are explored by the authors of [83,84]. Structured methodologies are proposed to assist specialist technologists in the decision-making process to solve these problems.

On the other hand, in [85] a specialized methodology for selection of technological processes in additive manufacturing is presented. The methodology is particularly useful in the early stages of design, allowing early identification and preventing potential errors. Despite its advantages, the methodology is unsuitable for traditional manufacturing processes, which are quite different from additive manufacturing.

Product design also has an important role to fulfill in successful manufacturing process [86–90]. Another source [91] presents a methodology for rational design selec-

tion that involves a purposeful sample survey. This methodology highlights the importance of integration between different aspects of the technological process, such as materials science, engineering and manufacturing technology. The application of the methodology requires significant resources, such as time investment for data collection and financial resources for conducting more in-depth research. However, it offers a summary result for the selection of a technological process, taking into account the many criteria considered.

According to the authors of [92], the AHP allows the different criteria to be systematized and to determine their relative weight in the decision-making process. The methodology enables objective comparison of several alternative concepts and assists teams in making strategic decisions related to lean manufacturing. The application of the methodology is intended to assist managers in deciding on a lean concept.

The methodology proposed by the authors of [93] for evaluating and modeling production processes focuses on improving their evaluation and management. With the help of information flows and interactions between different departments and individual stages of production as well as by following the steps of the proposed methodology, an in-depth analysis can be performed with the improvement in production processes. The methodology has great potential, but its complexity is a major obstacle for its implementation in the production process. This will have a particular impact on smaller enterprises or organizations with limited technological resources, where the implementation of such complex systems may prove difficult to achieve in practice. This requires qualified staff and investment in training and infrastructure, which is not always possible for smaller enterprises.

The methodology proposed by the authors of [94] evaluates different technological processes for a workpiece that is produced by casting. For the evaluation of technological processes, the authors compare different MCDM methods (TOPSIS, VIKOR and AHP) and analyze the most suitable method for application. In the same application area, authors of [95–99] proposed different methodologies for the selection of metal cutting tools with more powerful MCDM methods. With the rapid development of technology, the number of different milling machines is also increasing, for this reason the authors of [84] consider the use of MCDM methods in the selection of machine tools and machine tools. Other authors [100,101] develop their own methodology for the selection of cutting parameters in machining using MCDM methods.

In the same context, the authors of [102] develop a method for the rational selection of machining tools using prediction procedures. Through the “Objective Tree”, the method is an innovative approach that creates a prerequisite for developing a competitive product. The selection of appropriate tools is facilitated by the proposed method but also encourages development in design and production.

The authors of [103] propose an advanced methodology for supplier selection in digital supply chains of e-commerce platforms, based on an extended VIKOR approach using interval-valued intuitionistic fuzzy numbers (IVIFNs). Their decision framework addresses the challenges of uncertainty and conflicting criteria in dynamic digital environments. The study highlights the increasing role of MCDM methods in supply chain optimization, particularly in e-commerce contexts where digital transformation requires adaptive and precise evaluation tools. This approach demonstrates the practical applicability of VIKOR in real-world industrial scenarios, complementing the broader use of MCDM methods in sustainable engineering and logistics.

Although there are a variety of methodologies and methods to technological process design and optimization, many have limitations that prevent their widespread application. While effective in certain areas, the methodologies considered often fail to provide a universally applicable solution for the following reasons:

- Most methodologies have been developed for specific types of processes or industries, which narrow their applicability to a wider range of manufacturing contexts.
- No methodology effectively integrates quantitative analyses with a comprehensive consideration of technical and economic aspects, which is essential for meaningful process evaluation.
- The dynamic nature of market requirements and technological advances requires methodologies that can adapt quickly to change. Many existing methodologies do not meet this requirement, which limits their effectiveness in the long term.

The need to balance different criteria, the limitations in resources and the specific needs of production require the development of more flexible and adaptable solutions. Methodologies that use multi-criteria analysis to support decision making on process selection across different industries and their aspects show the potential for innovative approaches that can be adapted to current and future market requirements. Such solutions need to be accessible and applicable to both large and small enterprises to ensure high quality of the final product and sustainability of the production process in the long term.

2.4. Patents Using MCDM Methods for Technological Process Evaluation

The invention in the patent [104], which is a multilayer digital factory with integrated MCDM, offers an innovative and detailed method to the digitalization of the manufacturing process that can significantly improve the efficiency, management and flexibility of enterprises. The patent uses a six-layer architecture combined with various simulation models to enable more efficient control and management of each stage of the manufacturing processes. The architecture analyzes multiple criteria by processing and integrating data from different production levels. Based on the considered criteria, the system used in the patent adapts its solutions to the implemented architecture. However, the successful implementation of such a system requires careful and detailed planning, especially in terms of integration, security and the resources required. The method described in the patent has the following main advantages:

- Before the actual application of the technological processes, they are optimized;
- The most effective solutions are identified through simulations and analyses;
- Costs and time associated with development and implementation are reduced;
- Production efficiency is increased and product quality is improved.

Increasing production efficiency, reducing operating costs and improving the quality of production are some of the parameters that the proposed method in the patent provides enterprises with the opportunity to choose the most appropriate technological process for. The built-in tools for simulation and analysis of production processes show great potential by using real production data. In this way, companies can make informed and justified decisions.

Using computational models providing detailed solutions, the authors of the patent [105] present a methodology for monitoring manufacturing processes. It incorporates a combination of input data selection and optimization techniques with a mechanism to evaluate the performance of the models used. Through the Mahalanobis distance, Zeta-statistics and mathematical tools and algorithms for optimizing input parameters, the reliability of the models is increased. These methods help in better understanding and control of the models in different industrial applications.

Another patent [106], which describes the horizontal structured modeling of manufacturing processes and uses logic algorithms and models to make multi-criteria methods, provides tools for virtual modeling, simulation and the association of manufacturing characteristics. The method aims to improve the efficiency of computer-aided design and

manufacturing (CAD/CAM) by simplifying the work with complex manufacturing models. Technologists and engineers can perform the following:

- Virtual modeling and simulation of production processes;
- Automated decision making and optimization;
- CAD/CAM integration with production processes;
- Adaptive and efficient process control.

This lays the foundation for smarter and well-founded production planning, which is essential for success in today's environment.

A method for selecting a differentiated secondary equipment transformation scheme is described by the authors in the patent [107] for selecting a differentiated secondary equipment transformation scheme. A structure consisting of a scheme layer, criteria layer, and target layer, based on the usage history of the secondary equipment and the associated specifications and standards, is created. A method for quantifying the indices using “fuzzy number theory” is applied. The weight of each index is determined by using the same theory. The method finds application in the selection of differential transformation schemes of low-voltage electrical equipment (used for monitoring, control and regulation).

The main objective of the patent presenting a quantitative assessment tool [108] is to propose a tool combining a theoretical MCDM method with standard life cycle analysis techniques and statistical methods to facilitate decision makers in resource management. The method of the patent under review applies to the MCDM method—the AHP. The underlying methodology used is based on the method itself. The method is used in the selection of a differential scheme for the transformation of low-voltage electrical equipment (used for monitoring, control and regulation). The main objective of the patent, which presents a quantitative assessment tool [108], is to offer a tool that combines theoretical MCDM methods with standard life cycle analysis techniques and statistical methods to facilitate decision making in resource management and justification. The method of the review patent applies to the MCDM method—the AHP. As the basic methodology that is used steps on the foundation of the method itself, the AHP method provides a systematic and precise pairwise comparison of individual criteria. This technique clearly identifies and quantifies the relative importance of each criterion, which creates a sound basis for decision making. The application of the patented method is particularly significant in the selection of differential transformation schemes for low-voltage electrical equipment used for monitoring, control and regulation. During the application process, detailed information is acquired on both the characteristics of the present system and the suggested alternative solutions. Each choice is evaluated against predetermined criteria on a scale of 1 to 10, yielding an objective and reliable quantitative rating. Because of its accuracy and transparency, the methodology is frequently employed in a variety of sectors, including operational management, strategic planning, resource allocation and risk management.

A system for calculating a universal sustainability index using political, economic, social, technological, legal and environmental (PESTLE) factors describes the patent [109] for data processing that evaluates different sites in order to create a universal sustainability index through data collection. The system collects and analyses data provided by participants through surveys, questions, tests and other sources to create a universal sustainability value or numerical index. The primary goal is to create a methodology for assessing the resilience of various entities against a given universal resilience program. The criteria as well and their weights, value ranges or other quantitative assessments are provided by the participants. Normalization of the data is performed from 0 to 100 and assigned to a positive or negative effect. Participants assign percentage weights to the different criteria that influence the “sustainability index”. A vote among the participants determines the final result. For each criterion, the weighted value is calculated, then the percentage weight

is multiplied by the normalized value. To generate an overall “sustainability index”, the weighted values for the different criteria are combined. Against a given adjective, the “sustainability index” considers different criteria through a complex and long process of data collection, information and its processing. The main disadvantage of the patent is that it uses multiple participants, which can lead to different estimates, given the opinions of the participants.

The patent [110] concentrates on the different customer service opportunities of the organization, providing recommendations on the different options. The creation of a method that is adaptable, repeatable and can simplify the analysis, the processing and the collection of data related to customer service is the main goal of the patent. The methodology in the patent uses computer generated surveys, rating formulas and graphs displaying weight scales. This provides an effective analysis of customer service capabilities. The main steps in the patent are as follows:

- Initial data is collected through interviews with executive directors, surveys, etc.;
- Complex criteria are applied to evaluate and assign values;
- Secondary data is collected and normalized for further processing by surveying subject matter experts (SMEs);
- The final step is to analyze and prioritize potential improvements using various analytical tools such as evaluation formulas.

The results are shown in graphs, reports or analyses. In this way, an overview of the areas of improvement in the organization is developed. The “total impact of customer service improvement” formula seems to be the dominating valuation within the patent. The patent describes processes and techniques to be able to optimize customer service while elaborating on the need for improvement to be prioritized in organizational capabilities.

The systems and methodologies outlined in these patents illustrate the increasing contribution of digitization and advanced analytical methods for future industrial businesses. Businesses can improve efficiency, reduce costs and increase competitiveness in a global marketplace by adopting new technologies.

Although this review did not strictly follow a PRISMA procedure, the approach to selection remained methodologically transparent and relevant.

3. Results

3.1. Analysis of MCDM Methods

Based on the analysis of the available literature, the main advantages and disadvantages of different MCDMs used in manufacturing environments and engineering applications are identified. The studied methodologies demonstrate significant differences in the way the criteria weights are defined, the level of complexity of their application and their adaptability to specific manufacturing conditions [111–114].

In this comparison table (Table 1), some of the most commonly used MCDM methods are presented and evaluated against objective, quantifiable metrics. The table includes metrics such as the number of pairwise comparisons required, calculational complexity, the presence of a consistency metric (Consistency Ratio), the ability to handle interdependencies and their applicability to multi-criteria evaluation [115–119].

The presented table provides an objective assessment of MCDM methods using measurable indicators for each individual method. This approach allows for a more in-depth and accurate analysis of the methods, facilitating the selection of the most appropriate method while minimizing the subjectivity typical of qualitative assessments.

The data in Table 1 suggest an important exchange: methods with few comparisons and low computational complexity (e.g., FUCOM, BWM) reduce the workload on experts but risk missing dependencies between criteria; conversely, the ANP addresses these

dependencies but at the cost of many comparisons and high complexity. The lack of a formal consistency metric in the TOPSIS, VIKOR and SAW does not automatically mean invalidity, but it does require compensatory measures (e.g., sensitivity of results to weights and normalization). Overall, the table supports the use of the FUCOM for multiple criteria and a limited time, but in the presence of strong backward dependencies between criteria, it is justified to prefer the ANP or a hybrid approach.

Table 1. Comparison table of commonly used MCDM methods evaluated against objective application criteria.

Method	Number of Comparisons	Computational Complexity	Consistency Ratio (CR)	Interdependence Processing	Multi-Criteria Applicability
AHP	Large number	Medium	Yes (CR < 0.1)	No	Moderate (Limited when many criteria are present.)
ANP	Very large number (method uses network structure)	High	Yes (CR < 0.1)	Yes	Medium (Complexity of method limits scale of applicability.)
FUCOM	Small number	Low	Yes (CR < 0.1)	Limited (no complex relationships considered)	High (Very good when large number of criteria are present)
SAW	Does not use direct comparisons (weights are set directly and not compared)	Very Low	No consistency reported	No	High (Good for simple problems with few criteria. Limited handling of interdependencies.)
BWM	Small number (comparison of best and worst only)	Low/Medium	Yes (CR < 0.1)	No	Medium (Suitable with a small number of criteria)
VIKOR	Does not use direct comparisons (uses distance calculation from ideal solutions)	Medium	No consistency reported	No	High (Suitable for situations with distinct differences between alternatives. Easy to apply when many criteria are present.)
TOPSIS	Does not use direct comparisons (based on close-to-positive and -negative ideal)	Medium	No consistency reported	No	High (Suitable for quantitative data and clearly defined criteria. Works well with a large number of quantitative criteria.)

A comparison of the advantages and disadvantages of the most commonly used MCDM methods is presented in Table 2. Their evaluation in the table is based on Table 1 and the in-depth comparative analysis of advantages and disadvantages extracted from the existing literature.

Table 2. Advantages and disadvantages of commonly used MCDM methods.

Method	Advantages		Disadvantages	
AHP	✓	Clear hierarchical structure for evaluating criteria;	✗	Subjectivity in determining criteria weights;
	✓	Widely used and well understood method.	✗	Limited ability to deal with dependencies between criteria;
			✗	A large number of comparisons are needed for a complex problems.

Table 2. Cont.

Method		Advantages	Disadvantages
ANP	✓	Allows analysis of relationships and feedback between criteria; Suitable for complex problems with multiple interactions.	✗ When two attributes are considered and compared for decision making, the different effects between clusters can be ignored;
	✓		✗ The network structure between attributes is complex, and different structures lead to different results;
			✗ To form a super matrix, a pairwise comparison of each attribute with all others is necessary.
SAW	✓	Easy to understand and implement; Suitable for quick decision making.	✗ Cannot handle complex dependencies between criteria and the results do not approximate the early situation;
	✓		✗ Results are highly dependent on weights, which are subjectively determined;
			✗ All criteria values must be positive and maximal.
BWM	✓	Ease of implementation and increased consistency in criteria weights; Reduces the number of comparisons to some other methods such as the AHP.	✗ In cases with many criteria or complex relationships between them, the method may produce incorrect results;
	✓		✗ As a fairly new method, the BWM has not yet been widely researched and applied, requiring more academic and practical research to confirm its effectiveness in different domains.
FUCOM	✓	Offers a high degree of consistency in determining the weights of criteria; Reduces errors that may arise from subjective evaluations of criteria; Saves time and simplifies the process by reducing the number of comparisons required by the decision maker; The method is used in various fields, and its universality is due to its easy integration with other tools and MCDM methods, making it relatively easy to understand and implement, suitable for a variety of practical applications.	✗ Difficult to define some criteria and dependence on expert judgements; ✗ Need to normalize the quantifiable criteria.
	✓		
	✓		
	✓		
VIKOR	✓	Obtaining a predetermined solution at the beginning of the development; The method provides a trade-off ranking of alternatives, taking into account both the best and worst criteria values.	✗ Inappropriate ranking of criteria may occur due to the use of the VIKOR equation to calculate maximum group utility;
	✓		✗ When alternatives have close scores, it may be difficult to make a clear choice without additional analysis or information;
			✗ The method assumes that the criteria are independent of each other, which is not always true in real-world situations.
TOPSIS	✓	A simple and logical process based on the “closeness” of the criteria to the ideal point (best criteria values) and the farthest point (worst values); The method is based on linear algebra and geometry, making it reliable and consistent.	✗ Does not take into account the relative importance of distances from the ideal solution;
	✓		✗ The presence of extremely high or low values can affect the final result;
			✗ The method works best with quantitative data and can be difficult to apply when the criteria are qualitative or subjective.

The disadvantages of the methods are reduced down to three recurring sources of risk: heavy reliance on expert assessments and subjectivity (AHP, BWM and FUCOM); assumption of independence between criteria, which is rarely valid in engineering systems (SAW, TOPSIS, VIKOR) and sensitivity to normalization and extreme values (especially TOPSIS).

Table 3 shows some of the most commonly used MCDM methods and their areas of application in different industrial sectors. The applicability of the methods is explained by analyzing the literature using the AHP, the ANP, the TOPSIS, SAW, etc., in areas such as manufacturing, energy, healthcare, etc. The table demonstrates how some of the methods find practical application in specific situations in each field.

Table 3. Application of MCDM methods in different industrial fields.

	Industry	Application
AHP	Mechanical Engineering	Selection and evaluation of additive manufacturing processes [120] Selection of suitable material for robotic arm frame [121] Selection of an industrial robot for milling applications [122] Selection of the most suitable machine tool for machining applications [95]
	Automotive	Model evaluation and selection in vehicle design [123] Evaluation of suppliers in the automotive industry [124] Selection of green suppliers in the automotive industry [44]
	Power Engineering	Evaluation and ranking of different energy efficiency measures [125,126]
	Healthcare	Evaluation and selection of medical therapies and technologies, taking into account effectiveness, cost and patient preferences [127] Selecting the most appropriate location for new hospitals [128]
	Other	Prioritization of strategic objectives and evaluation of alternative strategies [129] Urban planning to prioritize projects based on social, economic and environmental criteria [130] Supplier evaluation [131] Identification of potential groundwater recharge zones [132]
ANP	Mechanical Engineering	Selection of optimal material, taking into account interdependencies between criteria [133] Selection of lean technologies in production chains, evaluating both qualitative and quantitative criteria [134]
	Automotive	Supplier selection in an automotive organization [135] Choosing green SCM strategies in an Indian automotive company [136]
	Transport and logistics	Evaluation and prioritization of sustainable sourcing practices (planning and control phases in manufacturing plants) [41] Selecting a logistics service provider [137]
	Other	Sensitivity analysis for planning the location of temporary facilities in construction projects [138] Supply efficiency assessment in the consumer electronics industry [139]
SAW	Mechanical Engineering	Optimization of the abrasive waterjet cutting process (determination of optimum parameters for cutting high-alloy steel) [61] Selection of a suitable robot [67] Selection of flexible manufacturing systems (FMS Flexible manufacturing systems) [67] Selection of non-traditional machining methods [67]
	Automotive	Selection of spare parts suppliers in the automotive industry [140] Automotive purchasing selection [141]
	Other	Selection of suppliers in a construction and logistics company [142] Selecting partners for reverse logistics in electronics [143] Selection of machines (for yogurt) in the food industry [144]
BWM	Mechanical Engineering	Process selection for additive manufacturing [145] Selection of machines and materials for additive manufacturing [146]
	Logistics	Selection of optimal transport mode for delivery of products to market [20]
	Automotive	Vehicle selection based on criteria set by the customer [20]
	Other	Supplier evaluation [20] Identification of potential groundwater recharge zones [132]

Table 3. Cont.

	Industry	Application
TOPSIS	Mechanical Engineering	Optimization of process parameters [147] Parameter optimization in additive manufacturing [148] Evaluation and selection of appropriate tooling [149] Determination of optimal parameters in milling [150]
	Automotive	Selection of brake materials [44] Choosing green suppliers in the automotive industry [45]
	Chemical Industry	Supplier selection in the oil and gas industry [151]
	Other	Evaluation and selection of initial training aircraft [19] Evaluation and ranking of transportation alternatives based on criteria such as cost, efficiency and environmental impact [152] Evaluation of suppliers [132,142]
FUCOM	Automotive	Selection of car brand [153] Evaluation of alternative vehicles [154] Sustainable urban mobility [155]
	Electronics and Electrical Engineering	Choosing a smartphone brand [156] Risk management in the electricity sector [157]
	Mechanical Engineering	Selection of suitable forklifts [26] Selection of automatic vehicles and equipment in warehouses [158,159] Selection of machinery and equipment for container handling [160] Selection of delivery vehicles [161] Selection of suitable painting robots [162] Multi-robot path planning in a cloud environment [163] Selection of non-traditional manufacturing methods [28] Hole turning [164]
	Agriculture	Sustainability assessment of village tourism sites [165] Site selection for biogas production [166]
	Energy and Power Engineering	Assessing the sustainability of oil supply chains [167] Selecting a suitable floating solar panel system [168] Selection of desalination and renewable energy power systems [169] Selecting the most suitable location for a biogas plant [165]
	Civil Engineering	Site selection for construction of single-panel bridges [170] Estimation of construction costs [171] Estimation of placement of facilities [172] Location selection for textile factories [173]
	Healthcare	Performance management in healthcare [174] Evaluation and selection of medical waste treatment method [175,176] Assessing sustainability in the healthcare sector [177] Supply management in pharmaceutical industry [178]
	Chemical industry	Resistivity in treatment of sewage sludge [179] Route planning for hazardous materials [180]
	Aircraft	Performance evaluation of airlines [181]
	Other	Human Resource Management [77,182] Supplier selection [30,183–186] Selection of transport [187] Logistics processes [188–190] Distribution [27] Business process management [191,192] Video streaming [193] Packaging recycling [194] Waste disposal [195] Green Innovation [196] National Parks [197] Mining mapping [198] Identification of potential groundwater recharge zones [132] Transport engineering and rail infrastructure [199–202] Reliability and risk in industry [203]

The flexibility of MCDM methods allows adaptation to the specific needs of manufacturing, energy, healthcare, finance, engineering and other industries, providing optimal weighting of multiple criteria and increasing the reliability of selections in complex environments.

The results of the analysis of the MCDM methods used in this study demonstrate their extraordinary value in manufacturing and engineering environments. The main advantages of these methods include their ability to process a large volume of data and take into account multiple criteria, including those with conflicting interests.

The ability of methods such as TOPSIS and VIKOR to effectively compare and select alternatives according to their proximity to the “ideal solution” and their distance from it, while simultaneously using multiple criteria, makes them widely used methods in industrial practice.

Easy and intuitive for users to use are MCDM methods such as SAW and the BWM, but they do not always manage to overcome the difficulty in complex relationships between criteria. In such cases, it is better to use a method that considers the interrelationships between criteria. Such an MCDM method is the ANP, but on the other hand it requires significant resources and computational time.

Methods such as BWM and FUCOM have low computational complexity compared to other MCDM methods. They are practical, require few pairwise comparisons and produce highly consistent results. FUCOM in particular shows exceptional efficiency when working with multiple criteria, making it very suitable for implementation in production environments, helping to optimize decision-making processes and achieve sustainable results.

The MCDM method, the FUCOM, differs from other methods in its significant potential to reduce subjectivity and increase consistency in determining the weights of the criteria and wide applicability, but at this point there are no sources confirming its application in the selection of technological processes, which is why greater attention is paid to the method and the study of its possibilities at this point. This makes it particularly suitable for applications that require high accuracy and efficiency.

The analysis highlights the variety of possible applications of the method, as well as its adaptability and effectiveness in different production and business environments. The examples presented show how the FUCOM supports strategic decision making by optimizing the selection of equipment, resources and processes. The comparison provides a clear insight into the potential of the FUCOM and identifies areas where it has proven its practical utility, facilitated its future applications and guides further research and practice. From the analyzed literature and the applicability of the method, it is clear that it is a preferred method for use in the field of mechanical engineering, being mainly used for the optimization and selection of machinery, equipment and robotic systems and logistic processes in manufacturing, but no applications of the method are found for process evaluation in mechanical engineering.

Figure 6 shows the percentage distribution of FUCOM applications in different industrial sectors. The largest share is occupied by applications in mechanical engineering and logistics, followed by the energy and healthcare sectors. These statistics show which industries are already benefiting most from the method and where there is potential for future expansion.

Despite the potential of the MCDM method—the FUCOM, its application in manufacturing environments, especially in mechanical engineering and specifically in technological process selection decisions, has not been studied sufficiently and is not represented in the available literature. That is why increasing the application of FUCOM in the field of mechanical engineering could make a significant contribution, thus providing new solutions and better results in the development of technological processes.

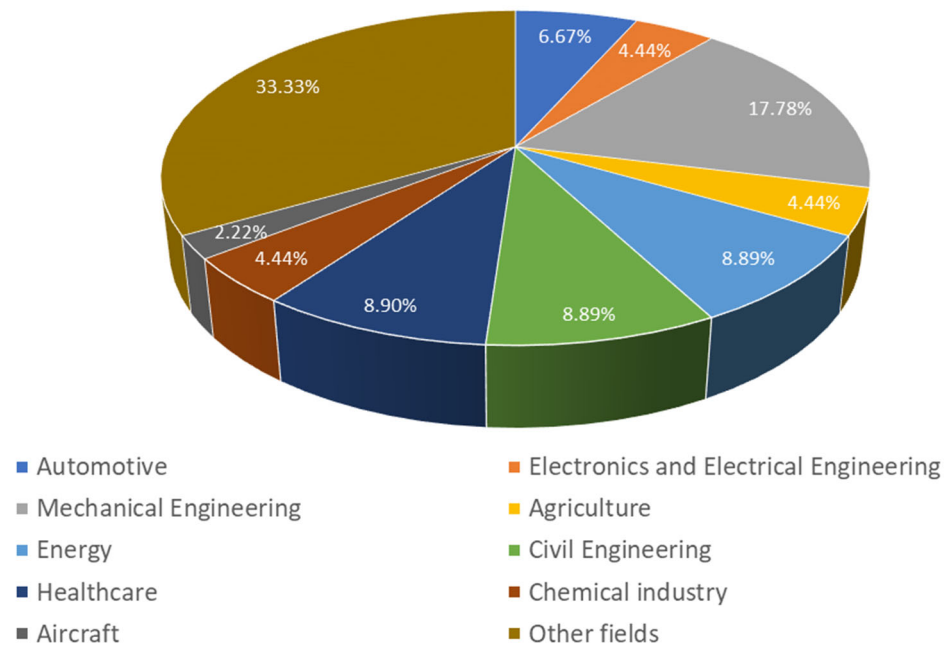


Figure 6. Percentage of FUCOM applications in the industry.

Based on the analysis carried out at the FUCOM point, although it is a relatively new MCDM method, it has proven its effectiveness in many industries. Thanks to its ability to reduce subjective errors and ensure consistency of results, its potential can also be exploited in the mechanical engineering sector. Focusing on the optimization or selection of suitable technological processes with the help of the FUCOM would represent an important contribution from both a scientific and practical point of view.

The data in comparative Tables 1 and 2 provide an objective assessment of the MCDM methods considered and present their advantages and disadvantages. Analysis from the tables can be extremely useful in selecting the appropriate method in specific engineering or manufacturing needs, thus an informed decision can be made.

3.2. Analysis of the Reviewed Patents and Methodologies Used

In order to further understand the relationships and analyze the strengths and weaknesses of each of the methodologies considered, Table 4 is presented, where the main characteristics and specific applications of each patent are highlighted, focusing on their advantages, disadvantages, their application and the MCDM method used.

The table includes detailed aspects such as the methods used, the MCDM methods in the patent methodology, the application, and the implementation process.

This comparative presentation allows an easy and systematic examination of each methodology, providing clarity on their applicability and innovation.

Similarities:

- All patents use different multi-criteria decision or evaluation methods or may adapt one in their methodology;
- All patents aim to improve existing processes, whether it is equipment optimization, improving sustainability, or prioritizing customer service;
- Each of the patents reviewed uses quantitative assessments, allowing informed decisions to be made and greater objectivity to be achieved in the overall analysis;
- The patents examined are designed to improve efficiency in various manufacturing sectors or industries;
- The patents contain clearly defined and structured decision-making processes.

Differences:

- Each patent is aimed at a different sector of application, making it difficult to implement in a different industry;
- Different methodologies and techniques are used;
- All patents require the participation of experts to evaluate the criteria under consideration.

Table 4. Comparison table of different patents using MCDM methodologies.

Patent	CN109298685 (A) [104]	US2006230097 (A1) [105]	US2004153200 (A1) [106]	CN110727912 (A) [107]	US20050114281 (A1) [108]	US20110093420 (A1) [109]	US8200527 (B1) [110]
Name	Method and system for digital factory implementation	Methodology and process model monitoring system	Horizontal structured modeling of the production process: an externally coupled representational embedding	Method for selecting a differentiated transformation scheme of secondary equipment	Quantitative assessment tool	Computer processing system for the assessment of resistance to PESTLE factors	Method for prioritizing and presenting customer service recommendations
Used methods	Six-layer architecture, simulation models for resources, processes and logistics	A logic module that processes multiple criteria	Logical and algorithmic models	FAHP	AHP	PESTLE analysis	Assessment Formulas, Weighing Goals and Complexity Levers
Data source	Data from production resources, their characteristics and operational data	Biometric sensors, personal information	Logical arrays, database structure	Historical information for the equipment	Collection and normalization of life cycle data	Data collection through network and expert evaluations	Computer data collection and survey questions
Application	Plants and factories aiming for digital transformation	Applicable in complex manufacturing systems	Automotive, robotics and automated manufacturing	Low-voltage electrical equipment industries	Industrial plants for monitoring and resource management and planning	Automation and centralized control of manufacturing processes	Contact center and customer service performance analysis
Implementation	Create simulation models related to manufacturing processes and logistics	Implementation of sensors and cryptographic technologies to protect personal and corporate data	Optimization of processing large amounts of data using algorithm	Requires detailed analysis of current equipment	Using mathematical models and financial information	Platform for multi-stakeholder participation	Contact center data entry, automatic analytics generation
Advantages	High efficiency, integration of existing systems, improved visibility	Data security, accuracy and integration	Increased speed of data processing, reducing errors in systems	More accurate transformation decision making	Accuracy and traceability of decisions	Involvement of experts and use of PESTLE factors	Automated data collection and processing
Disadvantages	Complexity of implementation in existing enterprises	Requires high initial investment and staff training	Complexity in tuning algorithms for specific cases	Use of sophisticated mathematical methods	Depends on extensive benchmarking	Requires multi-stakeholder participation for data collection	Multiple questions and instances required for accurate analysis
Year	2019	2006	2004	2020	2005	2011	2012
Country	China	USA	USA	China	USA	USA	USA

The patents reviewed so far and the various data processing, modeling, optimization, selection, etc., methods they use highlight the different industries in which they are used. Through the use of complex mathematical models for the FAHP method, the authors of patent [107] provide reliability in technical contexts. The patent [104], focused on architecture and the digitalization of the manufacturing process, is adaptable to the conditions of the modern industry. Patent [110] utilizes survey research, strategy, and weightings to provide better quality customer service and improved organizational processes. MCDM

methods, characterized by their flexibility and adaptability, demonstrate their ideal utility for application in volatile production environments. Through quantitative analysis with nonlinear regression, patents such as [108] estimate cost and efficiency. For the observation in adaptive systems, patent [105] uses dynamic monitoring. For the robustness of the methodology used in the patent [109], a PESTLE analysis is applied, which goes beyond purely technical frameworks, focusing on socio-ecological aspects.

Table 5 provides a comparative analysis of the patents examined based on a quantitative assessment. A direct comparison between them cannot be made because of the differences in the fields of application as well as the methods they use. For the purposes of Table 5, a focused assessment of the patents under consideration was carried out using a developed internal methodology. Each patent was assessed according to four predefined criteria: consistency, subjectivity, robustness and flexibility. The aim is to provide a structured and objective comparison of patents in the context of their applicability in an industrial environment.

Table 5. Comparison table of patents using MCDM methods, based on quantitative evaluation.

Patent	Consistency	Subjectivity	Sustainability	Flexibility	Summative Assessment
CN109298685 (A) [99]	8	7	8	9	80%
US2006230097 (A1) [100]	8	8	9	8	82.5%
US2004153200 (A1) [101]	7	6	7	7	67.5%
CN110727912 (A) [102]	9	8	9	7	82.5%
US20050114281 (A1) [103]	9	8	8	7	80%
US20110093420 (A1) [104]	7	5	7	10	72.5%
US8200527 (B1) [105]	7	5	6	8	65%

A total of eight experts participated in their evaluation—two representatives from two different universities and two mechanical engineering companies. All participants have professional and academic experience in the field of mechanical engineering, the application of MCDM methods and working with patents. The final evaluation for each patent is calculated as the average value of all individual evaluations submitted.

The participation of experts from various institutions and companies aims to reduce the risk of subjective assessment and achieve a more realistic assessment of the practical applicability of patents.

Each expert evaluates each patent according to the four criteria listed, using a scale from 1 to 10, where 1 is the lowest score and 10 is the highest. A brief description and assessment guideline are provided for each criterion:

1. Consistency

It evaluates how logical and mathematically consistent the method described in the patent is and whether there are clearly defined steps and well-founded decision-making logic.

- 1–3: Unclear methodology, lack of logical structure;
- 4–6: Partial consistency; some stages are well justified;
- 7–9: High degree of logical consistency and structure;
- 10: Excellent consistency, with a clear and validated algorithm or model.

2. Subjectivity

It assesses the extent to which the results of the method depend on subjective judgments. Lower subjectivity is preferred.

- 1–3: The method is highly dependent on subjective opinions, with no control or verification;
- 4–6: Some subjective elements are present but partially controlled;
- 7–9: Limited subjectivity, with ways to reduce its influence;
- 10: Almost entirely objective approach with minimal human intervention.

3. Robustness

It assesses the extent to which the method can provide reliable results even with incomplete or partially inaccurate input data.

- 1–3: Results change dramatically with small changes in data;
- 4–6: Moderate sensitivity to inaccuracies;
- 7–9: The method shows stability in different scenarios;
- 10: High level of stability and reliability of results.

4. Flexibility

It assesses the method's ability to adapt to different production conditions, industries or types of problems.

- 1–3: The method is only applicable in a very narrow context;
- 4–6: Partially adaptable to new conditions;
- 7–9: The method can be applied in a variety of industrial environments;
- 10: Excellent versatility and easy adaptation to different situations.

The comparison between the patents is based on the cumulative score obtained from the weight of the criteria listed in Table 5.

Based on the analysis, the patents with the highest scores show a high level of consistency, low subjectivity and excellent adaptability to different conditions. The methods in the patents that receive an average score, such as [104,108,109], offer balanced performance, but further optimizations may be required to increase their robustness and consistency. Those receiving lower scores such as [106,110] need improvements in reducing subjectivity and increasing reliability when the decision maker is working with incomplete information.

From technical optimization and production processes to social sustainability and customer-oriented solutions taking root, a major difference in the patents and methods is considered and therefore, are in their specific orientation. No universal methodologies have been found so far that can be applied in different industries, to have the possibility to apply different methods and approaches. The application of the patents under review are specifically tailored to the particular needs of different industries and situations.

The MCDM methods and methodologies applied in these patents focus on the development of special tools and systems based on the importance of a systematic and structured approach to the selection of technology and other solutions.

This review is intended to assist practitioners and researchers in understanding how these methods can be applied to improve processes in their areas of interest, as well as to help identify appropriate solutions to specific problems.

4. Discussion

4.1. Discussions

The numerous MCDM methods and related patents discussed highlight their importance in modern industries. These methods not only offer effective solutions to complex problems, but also provide a basis for innovation, especially when integrated with new

technologies related to Industry 4.0. Even though there have been a lot of research on MCDM methods and how they are used in different industries, there is not much information on how they are used to track and optimize technological processes in mechanical engineering. Their integration into software programs/systems or applications that facilitate selecting a suitable or rational technological process is not found in the literature reviewed. MCDM methods, characterized by their flexibility and adaptability, demonstrate their ideal utility for application in volatile production environments. Indeed, combining flexibility for adaptation with automated systems, especially in production contexts where automation and speed of production are paramount, will allow for faster, informed and dynamic decision making.

The comparison of the analyzed MCDM methods shows that the selection of an appropriate method depends on the structure and complexity of the problem under consideration and boils down to four main criteria:

- **Criteria structure.** In the presence of hierarchical, independent criteria, use of the AHP is recommended. The FUCOM, SAW and the BWM use a considerably simpler structure for the criteria, where the relationships between criteria are “crossed” or there are inverse relationships. The appropriate method is the ANP as it alone allows for network modeling.
- **Number of comparisons.** When a small number of comparisons are needed, the FUCOM and BWM are recommended. Where weights need to be assigned in the evaluation process, the AHP, ANP, FUCOM and BWM are appropriate. For SAW, known weights are required in advance.
- **Presence or absence of predefined weights.** When weights are already available, SAW performs direct ranking, and when weights are not available, the AHP, ANP, BWM or FUCOM should be selected.
- **Scaling analysis of the set of alternatives.** When there are many different alternatives/criteria, methods such as SAW, the BWM and the FUCOM offer quick and significantly easier calculations compared with others, which makes them more suitable. When comparing a small number of criteria, the AHP and ANP methods can be chosen because the number of comparisons is manageable.

The specific characteristics and structure of each method are of critical importance when selecting an appropriate MCDM method. Table 6 presents some of the most commonly used MCDM methods, providing a summary overview of their strengths and weaknesses.

Table 6. A supporting table on the strengths and weaknesses of some MCDM methods.

	AHP	ANP	FUCOM	SAW	BWM	TOPSIS
Hierarchical structure	✓	-	✗	✗	✗	✗
Dependencies	✗	✓	-	✗	✗	✗
Small comparisons	✗	✗	✓	-	✓	-
High consistency	✓	✓	✓	✗	✓	✗
Easy application	-	-	✓	✓	-	✓
Many criteria	-	-	✓	✓	✓	✓
Fast results	-	✗	✓	✓	✓	✓

✓ Yes/Major force; ✗ None; - No/Not major force.

The patents, methodologies and MCDM methods reviewed so far offer specific solutions that improve upon existing methods. In doing so, they highlight the importance of innovation in extending the scope and practical application of MCDM methods. They successfully integrate with the specific needs and requirements of different industries; however, there is no real application for real-time process monitoring and control.

In order to unlock the full potential of MCDM methods, there is a need to provide easier data entry, including the development of software programs and applications with intuitive interfaces to support the end user. This will not only support the selection of real-time technology solutions but also make MCDM methods and related methodologies a trend as they are integrated into modern technology.

Despite the many advantages of MCDM methods, they also have their limitations, and these are related to the complexity of implementation, the need for experts and their evaluation criteria and significant resources for data collection and processing. All of the above may limit their applicability in smaller or resource-constrained enterprises.

4.2. Future Directions and Challenges

The development of new standardized methodologies, as well as intuitive software to facilitate the implementation of MCDM methods, will help to overcome these limitations. To reduce the need for expert judgment and increase the efficiency of MCDM methods, the creation of hybrid models that automatically adjust criteria weights would assist with this process.

The analysis performed so far in this paper leads to interesting perspective possibilities for future research in the field of MCDM methods. The integration of IoT platforms to process and provide updated data and the development of software programs and mobile applications related to them is one of the directions for the development of dynamic MCDM methods that can adapt in real time to the changing conditions of the manufacturing environment in mechanical engineering and especially for the selection of optimal and rational technological solutions.

Building upon these directions, another important avenue for advancing MCDM research lies in exploring hybrid approaches that combine the strengths of different methods. Such approaches are increasingly being recognized as particularly valuable in the context of Industry 4.0, where decision-making environments are complex, uncertain, and highly dynamic [204].

Recent research has also emphasized the promise of hybrid MCDM approaches in addressing the challenges of Industry 4.0 adoption. For example, ref. [205] proposed a combined FAHP—TOPSIS framework to evaluate the influence of Industry 4.0 technologies on manufacturing strategies. These insights illustrate how hybrid decision-making models can provide structured and reliable guidance for prioritizing emerging technologies in future manufacturing environments.

In addition, ref. [206] illustrates how hybrid fuzzy-based MCDM techniques can effectively guide practitioners in overcoming barriers to Lean Six Sigma 4.0 integration and provide a structured roadmap for sustainable operational excellence.

Increasingly, studies emphasize the need for context-sensitive, multidisciplinary frameworks that combine hybrid/fuzzy MCDM with stakeholder co-creation to guide SME innovation and technology choice decisions within the ongoing transitions to Industry 4.0 [207].

This is also the place to note the role of artificial intelligence (AI) in MCDM methods. While AI models have significant potential to support MCDM processes by processing large volumes of data in real time, their direct integration into industrial environments poses serious challenges. This is due to the fact that decision making is in the hands of experts, and current AI systems are primarily oriented towards optimization according to selected criteria (e.g., cost effectiveness and/or performance efficiency) but rarely offer mechanisms for automated compliance with industrial standards such as the ISO, ASME or Machinery Directive 2006/42/EC [208], as well as the future Machinery Regulation 2023/1230 [209]. The lack of guaranteed traceability and regulatory validation makes it difficult to apply

them to MCDM in critical processes, critical components and processes and components subject to compliance and/or validation.

A major issue is that AI models, including large language models (LLMs), often function as a “black box”—meaning that it is very difficult to explain why a particular alternative has been chosen. This is in total contradiction with the requirement for transparency and traceability of decisions set out in the current Machinery Directive 2006/42/EC and future regulatory frameworks of Machinery Regulation 2023/1230. Such systems can offer mathematically optimal solutions, but without any guarantee of safety, sustainability or compliance with regulations, directives and standards.

In parallel, scholars have also begun to explore alternative approaches, among which the FUCOM has attracted attention due to its specific advantages over traditional MCDM techniques. The FUCOM has not been used in mechanical engineering manufacturing as a process evaluation tool. It has some major advantages over other MCDM methods, such as the following:

- Consistency;
- Working with multiple criteria;
- Minimizing the number of errors;
- Universality;
- Small comparisons between different criteria;
- Easy application.

With these advantages, the FUCOM offers the potential for effective application in mechanical engineering production as a process evaluation tool. As mentioned, FUCOM is a relatively new MCDM method, and despite its low calculation complexity and integration, the creation of dedicated software tools, programs and applications to tailor the process to the specific needs of machine manufacturing are needed.

The FUCOM can serve as a solid foundation for building innovative, adaptable and intelligent solutions to meet the increasing demands for efficiency, sustainability and flexibility of manufacturing systems. In this way, the method can successfully overcome some of today’s most serious manufacturing challenges.

5. Conclusions

MCDM methods allow for balanced and reasoned decision making based on various criteria related to the problem under consideration. These methods are well established as an important tool for overcoming the complexity of today’s technological manufacturing challenges by considering the importance of criteria such as efficiency, cost, quality, sustainability and innovation. Choosing the right MCDM method and implementing them properly in a particular situation or case is of major importance since it can result in an apparent boost in efficiency, optimization of resources used, cost reduction and improvement in general process efficiency.

With the implications of these methods combined with modern technology, new methods or hybrid ones will be able to arise. Such technologies could include artificial intelligence (AI), Big Data, deep learning, Internet of Things (IoT), etc. While IoT provides the ability to continuously monitor systems and automatically modify or change processes, artificial intelligence can analyze vast amounts of data, in real time.

All of this would greatly increase operational efficiency, reduce error risk and improve decision-making accuracy. With the increasing digitalization of the modern industrial environment, concepts related to digital factories and simulation models, mentioned in the patents under review, guide the future of MCDM methods. To combine efficiency and innovation and at the same time keep the methods accessible to a wide range of

industrial applications, it is necessary that their development follows the guideline along with dynamic and intelligent systems that can adapt in real time.

There is a clear need for expert judgment when selecting and ranking criteria, especially in situations where conditions change. But in practice, using MCDM methods in manufacturing is not always straightforward. This shows why practical, standardized methods are important—they make it easier for companies of all sizes to apply them effectively.

The methods reviewed here follow different ideas. Each one has its own logic for deciding which criteria matter more and how they should be evaluated when decisions are complex. There is not a single approach that works for all situations, and that variety can be both a strength and a challenge.

One of the main advantages of these methods is how widely they can be used. They have been applied in areas like production digitalization, sustainability efforts and new ways of managing resources. They also work in settings that involve automation, smart management systems or decisions planning. Because of that, they offer tools that can fit different company goals.

Nevertheless, many of the methods have notable limitations. Some are hard to apply without expert help. Others take time, data or technical resources that are not always available. That makes the case for more research and more consistency in how these tools are developed and shared.

Early involvement of experts and availability of intuitive interfaces significantly improve the usability of MCDM tools for both engineers and decision makers. MCDM methods have other challenges to face besides the ones shown so far. An example of such difficulties is the limited versatility of some of the methods, which do not adapt easily and quickly to changing conditions, and the complexity of their integration process in small- and medium-sized enterprises. These challenges need to be overcome in order to increase efficiency and practicability.

Combining different methods, by creating hybrid variants of the original MCDM methods, can help overcome these obstacles and even extend their practical application.

The analysis of the considered MCDM methods showed that the optimal choice of a method depends on the particular industry, its specifics, the data features and their characteristics. The FUCOM is used in multiple industries with different applicability. In situations with many criteria and limited time for expert comparisons, the FUCOM provides reliable weights with minimal effort, which is an advantage in strategic planning and automated systems. Its use in manufacturing environments, especially in mechanical engineering, is still limited; no evidence of its application in technological process evaluation has been found, which raises the interest of the scientific audience in this direction and highlights the need for further research and its wider application in the future. In this regard, the current and future research of the research team is aimed precisely at the development of methods for its application in industrial practices, especially in the field of technological processes in mechanical engineering.

When a large number of alternatives need to be quickly ranked based primarily on quantitative metrics (e.g., supplier selection, evaluation of energy source options), the TOPSIS features high speed and intuitive interpretation of results. For detailed structured hierarchical problems supported by experienced experts—for example, selecting a manufacturing technology that combines technical, economic and environmental subobjectives—the AHP remains the best solution due to its transparency and consistency control. When medium or highly standardized criteria sets are available, MCDM methods such as the BWM and SAW are preferred because they offer more economical solutions. In cases where there are significant correlations between criteria, the choice of the ANP is recommended.

MCDM methods are not just academic ideas. As shown in this review, they are practical tools that help with selecting and improving processes in real industrial settings. They play a role in shaping the priorities of modern companies—things like innovation, sustainability and staying ahead of the competition. These objectives are grounded in practical decision-making processes rather than theoretical constructs, and MCDM methods help structure those decisions in a useful way. Because of that, we are likely to see these methods built into more software products. They will not just be used for solving difficult case studies or modeling complex scenarios. They will also support organizations in making smarter choices and holding on to their competitive position over time.

Future research of the scientific team will be oriented towards the development of hybrid methods that integrate the strengths of different MCDM methods, the creation of software platforms for complex calculations and exploring the capabilities of the FUCOM in process selection and evaluation. Future work will focus on integrating the FUCOM into real-time decision support systems, such as modular software or mobile applications. Pilot implementations in industrial settings are necessary to validate its utility and ease of use under dynamic conditions.

In line with the findings of this review, the authors have previously developed a structured methodology for the selection of rational technological processes, which is currently being implemented in actual production, and the results are yet to be published. As a continuation of this research, a software application is currently under development to implement this methodology in practice. The proposed solution is built upon the principles of multi-criteria decision making, with the FUCOM serving as the core evaluation and ranking mechanism. This integration aims to provide decision makers with a practical and adaptable tool for use in real industrial environments.

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Abbreviations

The following abbreviations are used in this manuscript:

MCDM	Multi-Criteria Decision Making
AHP	Analytic Hierarchy Process
FAHP	Fuzzy Analytic Hierarchy Process
ANP	Analytic Network Process
Fuzzy AHP	Fuzzy Analytic Hierarchy Process
TOPSIS	Technique for Order Preference by Similarity to Ideal Solution
BWM	Best Worst Method
SAW	Simple Additive Weighting Method

VIKOR	Višekriterijumsko Kompromisno Rangiranje
FUCOM	Full Consistency Method
DEMATEL	Decision-Making Trial and Evaluation Laboratory
AI	Artificial Intelligence
LLMs	Large Language Models
ISO	International Organization for Standardization
ASME	The American Society of Mechanical Engineers

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